

Advanced Macroeconomics

Bewley-Huggett-Aiyagari-Imrohoroglu Model

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References

- Ljungqvist-Sargent (Ch. 17 and some parts of Ch. 16): Textbook treatment.
- PhD Macro book (Ch. 11, consumption chapter written by Gianluca Violante)
- Aiyagari (1994), Huggett (1993): Original papers. Relatively easy to follow.
- Guvenen (2011, Macroeconomics with Heterogeneity: A Practical Guide): Comprehensive review starting from incomplete markets to model and extensions.
- Heathcote, Storesletten & Violante (2009, Annual Review): Overview paper without equations.
- Cherrier, Duarte, & Saïdi (2023, European Economic Review): History of household heterogeneity in macroeconomic models.

This lecture is a mix of the first three bullet points.

Introduction

Goal:

- Present the canonical dynamic general equilibrium model of incomplete markets with household heterogeneity.
- The framework is used to analyze questions such as:
 - ▶ How much of the wealth inequality can be explained by earnings variation across agents?
 - ▶ What are the distributional implications of various fiscal policies? How are inequality and welfare affected by such policies?
 - ▶ What are the macroeconomic consequences of this heterogeneity in aggregate variables and prices?
- We focus on the stationary equilibrium, the equilibria with **constant prices** through time.

Introduction

Model Ingredients:

- Typical consumption-savings problem in infinite horizon.
- Two important features:
 1. **Idiosyncratic Shocks**: Individuals receive exogenous “income shocks”: e.g., unemployment shocks, promotions, etc.
 2. **Incomplete Markets**: They cannot trade assets (there is no way to buy insurance in the market).
- There is **NO** aggregate uncertainty.

Introduction

- Individuals are *ex-ante* homogeneous.
 - ▶ ...but will be *ex-post* heterogeneous!
- Exogenous earnings distribution, but endogenous wealth distribution.
- **Intuition:**
 - ▶ Lucky individuals that receive a sequence of high-income shocks will accumulate assets to insure themselves against future low-income;
 - ▶ Unlucky individuals that receive bad shocks will have no assets;
 - ▶ Equilibrium will feature a non-degenerate stationary wealth distribution.

To fully solve the model, we go through three building blocks:

1. The household consumption-savings problem (asset supply function);
 - ▶ Solve the household problem;
 - ▶ Solve for the endogenous stationary distribution;
 - ▶ Use the distribution and the HH decisions to get the aggregate asset supply.
2. Asset demand function;
 - ▶ It can be from the aggregate production function (e.g., firms) or government;
3. Finally, find the equilibrium in the asset market;

Individual's Problem

- Discrete time, infinite horizon, future utility is discounted by $\beta \in (0, 1)$.
- Continuum of individuals with unitary mass.
- Earnings are given by $w_t s_t$, where w_t is the market wage and s_t is a labor endowment, which is idiosyncratic and follows a Markov chain with transition probabilities:

$$\pi(s', s) = \Pr(s_{t+1} = s' \mid s_t = s). \quad (1)$$

- The individual supplies labor inelastically. The per period utility function is given by: $u(c_t)$, where $u' > 0$, $u'' < 0$ and $c_t \geq 0$.

Individual's Problem

- Agents only have access to a riskless bond that pays an interest rate r .
 - ▶ No access to a full set of state-contingent Arrow-securities. This is the incomplete market.
- They can save and borrow, but there is a borrowing constraint ϕ .
 - ▶ We will assume that $\beta(1+r) < 1$ and will verify that this is the only equilibrium later.
- Full individual problem:

$$\begin{aligned} \max_{c_t, a_{t+1}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{subject to} \quad & c_t + a_{t+1} = w_t s_t + a_t(1+r_t), \\ & a_{t+1} \geq -\phi \quad \text{and} \quad c_t \geq 0 \quad \text{for } t = 0, 1, \dots, \infty \\ & a_0 \text{ is given.} \end{aligned}$$

- We will look for a stationary equilibrium so ignore time subscripts in prices for a moment (more on that later).

Borrowing Constraint

- The borrowing constraint can be set exogenously or be bounded by the **natural debt limit**.
- The natural debt limit is the maximum borrowing that the household can pay back (if $c_t = 0$ and s_{min} in all periods).
- Iterating forward:

$$c_t = ws_t + a_t(1+r) - a_{t+1} \geq 0 \Rightarrow a_t \geq -\frac{ws_t}{1+r} + \frac{a_{t+1}}{1+r}$$
$$a_t \geq -\frac{ws_t}{1+r} + \frac{a_{t+1}}{1+r} \geq -\frac{ws_t}{1+r} + \frac{1}{1+r} \left(-\frac{ws_{t+1}}{1+r} + \frac{a_{t+2}}{1+r} \right) \geq \dots$$
$$a_t \geq -\left(\frac{1}{1+r} \right) \sum_{j=0}^{\infty} \frac{ws_{t+j}}{1+r}$$

note that because $r > 0$ and a_{t+j} bounded, the limit of $a_T/(1+r)^T$ goes to zero as $T \rightarrow \infty$.

Borrowing Constraint

- The worst case scenario is when the agent receives the lowest realization in every $t + j$: $s_{min} = s_{t+j}$. Substituting and we get the natural debt limit:

$$a_t \geq \frac{ws_{min}}{r}. \quad (2)$$

- **Inada Conditions:** with Inada conditions ($u(0) = -\infty$), the consumer will never borrow up to the natural debt limit since this implies zero consumption.
 - ▶ The constraint acts as a “repelling boundary”.
- That is NOT true with ad-hoc borrowing limits above the natural one!
- Let us now consider the possibility that the borrowing constraint can bind.

Consumption-Savings Problem

- Consider the Karush-Kuhn-Tucker of the consumption-savings problem and let μ_t be the multiplier of the borrowing constraint.

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \{ \beta^t u(c_t) + \lambda_t (ws_t + a_t(1+r) - c_t - a_{t+1}) + \mu_t (a_{t+1} + \phi) \}$$

with KKT conditions $\mu_t \geq 0$ and $\mu_t (a_{t+1} + \phi) = 0$.

- The solution implies the Euler Equation for all t :

$$u'(c_t) = \beta(1+r)\mathbb{E}_t[u'(c_{t+1})] + \mu_t$$

- If the constraint does not bind, $a_{t+1} > -\phi \Rightarrow \mu_t = 0$, we have the standard Euler Equation.

Consumption-Savings Problem

- If the borrowing constraint is binding, $a_{t+1} = -\phi$ and $\mu_t > 0$:

$$u'(c_t) > \beta(1+r)\mathbb{E}_t[u'(c_{t+1})].$$

- That means marginal utility of consumption at t is too high (i.e., c_t is too low). The household would like to consume more and smooth consumption but cannot do it.
- In this case, the household will just consume everything and hope for a higher income in the future.
- This situation may arise if the household is too poor (low wealth or low income) and/or the borrowing constraint is too tight. **Aiyagari (1994)** summarizes in a figure.

Optimal Savings

- Recall the Euler equation: $u'(c_t) = \beta(1 + r) \mathbb{E}_t[u'(c_{t+1})] + \mu_t$.
- Three reasons to save:
 1. **Intertemporal substitution:** trade-off between β and $(1 + r)$.
 2. **Consumption smoothing:** desire to smooth income shocks over time (e.g., saving for retirement; borrowing against an expected future raise).
 3. **Precautionary savings:** build a buffer against future uncertainty.
- Motives 1. and 2. are present even without income risk.
- Motive 3. requires either a special shape of $u(\cdot)$ or a binding borrowing constraint — this is central to the Aiyagari model.

Precautionary Savings

- Consider 2 periods with $\beta(1+r) = 1$, $a_0 = 0$, and $a_2 = 0$.
- **Deterministic income** $s_1 = \bar{s}$. The Euler equation gives:

$$u'(\underbrace{ws_0 - a_1}_{c_0}) = u'(\underbrace{a_1(1+r) + w\bar{s}}_{c_1}) \implies c_0 = c_1.$$

The only savings motive is if ws_0 differs from the desired c_0 — no extra buffer.

- **Stochastic income** $s_1 = \bar{s} + \varepsilon$, $\mathbb{E}[\varepsilon] = 0$. The Euler equation becomes:

$$u'(c_0) = \mathbb{E}[u'(\underbrace{a_1(1+r) + w\bar{s} + w\varepsilon}_{c_1})].$$

- **Problem:** $\mathbb{E}[u'(c_1)] \neq u'(\mathbb{E}[c_1])$ in general — we cannot conclude $c_0 = c_1$!
- Whether $\mathbb{E}[u'(c_1)]$ is larger or smaller than $u'(\mathbb{E}[c_1])$ depends on the **convexity of marginal utility** — Jensen's inequality.

Digression: Jensen's Inequality

- A function $f(x)$ is **convex** if, for $\alpha \in [0, 1]$ and two points x_1, x_2 :

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2).$$

- Consider a random variable x that equals x_1 with probability p and x_2 with probability $1 - p$:
 - ▶ Expected value: $\mathbb{E}[x] = px_1 + (1 - p)x_2$.
 - ▶ By convexity of f : $f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]$.
- For the general case, this result is known as **Jensen's Inequality**.
- In our case, whether $\mathbb{E}[u'(c_1)] \leq u'(\mathbb{E}[c_1])$ depends on the convexity of **marginal utility** $u'(\cdot)$.

Convexity of Marginal Utility

- Marginal utility $u'(c)$ is convex in c if and only if $u'''(c) > 0$.
- If $u'''(c) > 0$, Jensen's inequality applied to $u'(c_1)$ gives:

$$\mathbb{E}[u'(c_1)] > u'(\mathbb{E}[c_1]).$$

- Going back to the Euler equation with $\beta(1+r) = 1$:

$$u'(c_0) = \mathbb{E}[u'(c_1)] > u'(\mathbb{E}[c_1]).$$

- Since $u'' < 0$, this implies $c_0 < \mathbb{E}[c_1]$: the agent saves *more* than the pure smoothing motive dictates.
- This extra saving is **precautionary saving**: a buffer built against future income risk.

Savings: Risk Aversion vs. Prudence

- **Risk aversion** ($u'' < 0$): curvature of $u(\cdot) \Rightarrow$ consumption smoothing.
- **Prudence** ($u''' > 0$): convexity of $u'(\cdot) \Rightarrow$ precautionary savings.
- **CRRA utility** $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$: $u'' = -\sigma c^{-\sigma-1} < 0$, $u''' = \sigma(\sigma+1)c^{-\sigma-2} > 0$. ✓
- **Quadratic utility** $u(c) = -\frac{1}{2}(\bar{c} - c)^2$: $u'' = -1 < 0$, $u''' = 0$. No prudence!

$\Rightarrow$ Under quadratic utility there are *no* precautionary savings.

- **Key insight for Aiyagari:** even without prudence ($u''' = 0$), precautionary savings arise if the **borrowing constraint can bind** — households save to avoid hitting the constraint.

Precautionary Savings: Borrowing Constraint

- Suppose there is a non-zero probability that in $t + 1$ the borrowing constraint will bind.
 - ▶ In this case, the individual will NOT be able to smooth consumption.

$$u'(c_t) = \beta(1 + r)\mathbb{E}_t[u'(c_{t+1})]$$

- Even if the borrowing constraint cannot bind in $t + 1$, it may bind in the future.
 - ▶ Precautionary savings depends on how likely the constraint binds (how tight ϕ is, the stochastic process of s_t , etc).
- This motive is present even if $u(\cdot)$ does not have prudence (quadratic utility).

Consumption-Savings

- To solve the full consumption-savings problem, we can use standard dynamic programming techniques.
- The Bellman equation:

$$V(a, s) = \max_{a' \geq -\phi} \{u((1+r)a + ws - a') + \beta \sum_{s'} \pi(s', s) V(a', s')\}$$

with the associated policy function $a' = g_a(a, s)$ ($c = g_c(a, s)$ is recovered using the budget constraint).

- Like Aiyagari, if s is iid we can also use a cash-on-hand formulation.

From Partial to General Equilibrium

- At this point, we have taken w and r as given and solved the **partial equilibrium** problem of the consumer.
- Now, we proceed to solve the general equilibrium: we must find the r such that the asset market clears.
- We focus on the **stationary equilibrium**: the aggregates such as total assets, and prices will be constant over time, but the individuals will move up or down the earnings and wealth distribution!
- The equilibrium will feature a **stationary distribution**: a time-invariant distribution that will replicate itself every period.

Stationary Distribution

- The household is characterized by their pair (a, s) . Let the joint distribution of types be $\lambda_t(a, s) = \Pr(a_t = a, s_t = s)$ (CDF: Λ_t).
- Given the distribution of agents $\lambda_t(a, s)$, how can we find $\lambda_{t+1}(a, s)$?
- Intuitively, a household (a, s) moves to the next state according to the optimal policy function and the exogenous Markov chain.
- Let $Q((a, s), \mathcal{A} \times \mathcal{S})$ be the probability that a household with state (a, s) transits to the set $\mathcal{A} \times \mathcal{S}$:

$$Q((a, s), \mathcal{A} \times \mathcal{S}) = \mathcal{I}\{g_a(a, s) \in \mathcal{A}\} \sum_{s' \in \mathcal{S}} \pi(s', s)$$

Stationary Distribution

- To get the next period distribution, we just need to apply the transition function Q to all the points of the distribution:

$$\Lambda_{t+1}(\mathcal{A} \times \mathcal{S}) = \int_{A \times S} Q((a, s), \mathcal{A} \times \mathcal{S}) d\Lambda_t$$

- The stationary distribution is the distribution that replicates itself for all $(a, s) \in A \times S$:
 $\lambda(a, s) = \lambda_t(a, s) = \lambda_{t+1}(a, s)$.
- **Intuition:** if we discretize the asset space, Q can be interpreted as a transition probability matrix of a Markov chain with state-space $A \times S$.

Intuition using a Discrete Distribution

- Suppose we discretize the distribution in two asset states and two income states.
 - ▶ An entry $\lambda_t(a_i, s_j)$ is the fraction of agents in state (a_i, s_j) .
- The matrix Q is the transition matrix that governs the fraction of agents in state $\lambda_t(a_i, s_j)$ that moves to all states of λ_{t+1} :

$$\underbrace{\begin{bmatrix} \lambda_{t+1}(a_1, s_1) \\ \lambda_{t+1}(a_1, s_2) \\ \lambda_{t+1}(a_2, s_1) \\ \lambda_{t+1}(a_2, s_2) \end{bmatrix}}_{\lambda_{t+1}} = \underbrace{\begin{bmatrix} Q_{1,1} & \dots & \dots & Q_{1,4} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ Q_{4,1} & \dots & \dots & Q_{4,4} \end{bmatrix}}_Q \underbrace{\begin{bmatrix} \lambda_t(a_1, s_1) \\ \lambda_t(a_1, s_2) \\ \lambda_t(a_2, s_1) \\ \lambda_t(a_2, s_2) \end{bmatrix}}_{\lambda_t}$$

- We know that under certain conditions, the Markov chain admits a unique stationary distribution $\lambda(a, s)$.

Stationary Distribution

Interpretation of the stationary distribution:

- The fraction of time that an infinitely lived agent spends in the state (a, s) .
- Fraction of households in the state (a, s) in a given period in the stationary equilibrium.
- The initial *distribution* of agents remains constant over time even though the state of the individual household is a stochastic process.

Asset Supply

- Once we have the stationary distribution, we can find the aggregate asset supply by summing the savings of all households.
- In other words, we must integrate the distribution to find the **Asset Supply Function**:

$$\mathbb{E}a(r) = \int_{A \times S} g_a(a, s; r) d\Lambda(a, s; r).$$

- Note the dependence of r through the savings policy function and the distribution.
 - ▶ Asset supply is increasing with the interest rate: $\uparrow r \Rightarrow \uparrow \mathbb{E}a(r)$.

Closing the Model

- To close the model, we must define the demand for assets in the economy. Two options:
 - ▶ **Huggett (1993)**: Credit economy. Some agents borrow, others will lend. The loan market clears when aggregate demand for loans is zero.

$$\int_{A \times S} g_a(a, s) d\Lambda = 0$$

- ▶ **Aiyagari (1994)**: Production economy. Firms demand capital to produce. Market clears when household savings equalize capital demand.

$$\int_{A \times S} g_a(a, s) d\Lambda = K$$

- We follow Aiyagari (1994) and assume an aggregate production function.

- Let the production function be $Y = F(K, N) = K^\alpha N^{1-\alpha}$, where $\alpha \in (0, 1)$.
- Capital depreciates at rate δ .
- Markets are competitive and the solution of the firm problem is standard (t is omitted):

$$w = \frac{\partial F(K, N)}{\partial N} = (1 - \alpha) \left(\frac{K}{N} \right)^\alpha$$
$$r + \delta = \frac{\partial F(K, N)}{\partial K} = \alpha \left(\frac{K}{N} \right)^{-(1-\alpha)}$$

- Tight connection between w and r through the capital-labor ratio:
 $\uparrow r \Leftrightarrow \downarrow K/N \Leftrightarrow \downarrow w$.

- Notice that labor supply is inelastic, so aggregate labor is given by the sum of all labor endowments in the economy.
- Let $\Pi(s)$ be the invariant distribution of the Markov chain. Aggregate labor supply is:

$$N_t = \sum_i s_i \Pi(s_i)$$

- **Example:** two state Markov chain with $s_1 = 1$, $s_2 = 2$ and symmetric transition matrix.
 $N_t = 1 \times 0.5 + 2 \times 0.5 = 1.5$.

Equilibrium Definition

A **stationary recursive competitive equilibrium** is a value function V ; policy functions for the household g_a and g_c ; firm's choice K and N ; prices w and r ; and, a stationary distribution Λ such that:

1. Given prices, the V , g_a , and g_c solve the household problem.
2. Given prices, K and N solves the firm's problem:
3. Given the transition function Q , the stationary distribution satisfies:

$$\Lambda(\mathcal{A} \times \mathcal{S}) = \int_{A \times S} Q((a, s), \mathcal{A} \times \mathcal{S}) d\Lambda$$

4. The labor market clears: $N_t = \sum_i s_i \Pi(s_i)$.
5. The asset market clears: $\int_{A \times S} g_a(a, s) d\Lambda = K$.
6. The goods market clears: $\int_{A \times S} g_c(a, s) d\Lambda + \delta K = F(K, N)$.

Existence of Equilibrium

- Focus on the asset market: with Cobb-Douglas, it is easy to see that wage is just a function of r .
- To find an equilibrium, we must show that the excess demand function intersects at zero.
 - ▶ Technically, we need to show that it is continuous and strictly monotone.
- **Capital Demand:** from the firm's problem, capital demand is

$$K(r) = \left(\frac{\alpha}{r + \delta} \right)^{\frac{1}{1-\alpha}} N,$$

if $r \rightarrow -\delta \Rightarrow K \rightarrow +\infty$; if $r \rightarrow +\infty \Rightarrow K \rightarrow 0$.

Existence of Equilibrium

- **Asset Supply:**

$$\mathbb{E}a(r) = \int_{A \times S} g_a(a, s; r) d\Lambda(a, s; r).$$

- The asset supply is bounded above by: $(1 + r)\beta = 1$.
 - ▶ Intuitively, $(1 + r)\beta = 1$ is the complete markets/nonstochastic steady state equilibrium.
 - ▶ Because of precautionary savings, for a given r , the asset accumulation must always be higher than the certainty case.
 - ▶ With uncertainty, if $(1 + r)\beta = 1$, the agent will accumulate assets to $+\infty$.
 - ▶ See Ljungqvist and Sargent for the full argument.

$$r \rightarrow \frac{1}{\beta} - 1 \Rightarrow \mathbb{E}a(r) \rightarrow +\infty.$$

General Equilibrium

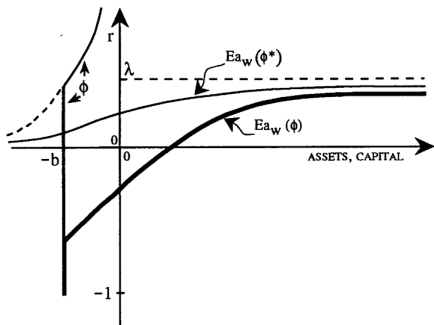


FIGURE IIa
Interest Rate versus Per Capita Assets

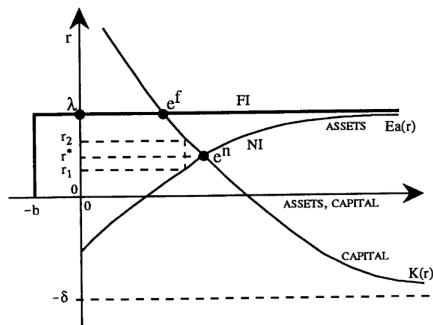


FIGURE IIb
Steady-State Determination

Source: Aiyagari (1994). Note: $\lambda \equiv \frac{1}{\beta} - 1$; $\phi^* \equiv \frac{ws_{min}}{r} > \phi$.

General Equilibrium

- In general equilibrium, r is determined endogenously by: $\mathbb{E}a(r) = K(r)$.
- Because of precautionary savings, aggregate savings will be higher than the case of certainty (and r will be lower).
- The tightness of the borrowing constraint, ϕ , is important. If agents are not allowed to borrow, precautionary savings will be higher and r will be even lower.

Quantitative Exploration

Calibration

- Most of the calibration is standard: $\alpha = 0.33$, utility is CRRA with γ between 1 and 5, $\delta = 0.08$.
- The labor endowment is an estimated AR(1) from a panel-data on labor income:

$$\log s_t = \rho \log s_{t-1} + \sigma \varepsilon_t,$$

where $\rho \in (-1, 1)$ and $\varepsilon \sim N(0, 1)$. Then discretize to a discrete Markov chain (more on that later).

- β is calibrated using information on the aggregate wealth-to-income ratio (K/Y).
- ϕ is calibrated to the fraction of agents with negative wealth (or assumed $\phi = 0$).

Aiyagari (1994):

- For reasonably calibrated parameters the differences between the savings rates with complete and incomplete markets are very small (at most 2%).
 - ▶ Although it can be much higher with $\uparrow \sigma$ and $\uparrow \rho$.
- Inequality follows qualitatively the same rank as the data:
 - ▶ inequality in wealth $>$ inequality in income $>$ inequality in consumption.
- Moreover, the distributions are right-skewed. Top-inequality is a feature of these models!
 - ▶ But the model was still a bit off quantitatively. Inequality was lower in the model.

Conclusion

- **Incomplete Markets Model:** new theoretical insights which open the door for old questions (capital taxation, government debt, etc).
- But, where the model shines is to provide a framework to study new questions related to income/wealth inequality.
- A large subsequent literature has worked to make the model match the distribution of wealth well.
- Then, study policies where inequality is central (progressive taxation, social security, etc).

Where to Go Now?

- **Progressive Taxation:** Boar and Midrigan (2022, JME); Kindermann and Krueger (2022, AEJ: Macro); Brüggemann (2021, AEJ: Macro); Heathcote and Tsujiyama (JPE, 2021); Conesa, Kitao and Krueger (2009, AER).
- **Hours and Labor Market Frictions:** Bick, Blandin and Rogerson (2022, QJE); Krusell, Mukoyama, and A. Sahin (2010, ReStud).
- **Top Wealth Inequality and Heterogeneous Asset Returns:** De Nardi and Fella (2017, RED); Benhabib, Bisin and Luo (2019, AER), Benhabib, Bisin and Zhu (2011, ECTA); Kaymak, Leung and Poschke (2022, WP).
- **Liquid and Illiquid Assets and the Wealthy Hand-to-Mouth (and MPCs):** Kaplan and Violante (2014, ECTA); Kaplan and Violante (2022, Annual Review).
- **Unsecured Credit and Consumer Default:** Chatterjee et al. (2007, ECTA), Livshits et al. (2007, AER); Dempsey and Ionescu (2022, WP); Herkenhoff (2019, ReStudies)
- **Non-Homothetic Preferences:** Straub (2022, WP); Mian, Straub and Sufi (2021, QJE); Carroll and Hur (2020, JME); Känzig (2022, WP).

Extension: Fiscal Policy

- **Precautionary Savings:** aggregate capital is higher than the Pareto optimal.
- In the baseline model, policies that **reduce** aggregate savings are Pareto improving. For instance: capital taxation and government debt.
- **Government budget constraint:**

$$G_t + (1 + r_t)B_t = B_{t+1} + T_t \quad \text{in SS} \Rightarrow \quad G + rB = T$$

where B_t is the government debt, G_t is the government consumption and T_t aggregate tax revenue.

- The market clearing conditions (in SS) also change:
 - ▶ Asset market: $\int_{A \times S} g_a(a, s) d\lambda \equiv A = K + B$.
 - ▶ Goods market : $\int_{A \times S} g_c(a, s) d\lambda + \delta K + G = F(K, N)$.

Extension: Fiscal Policy

- Suppose all households are subject to the same tax rates. HH budget constraint:

$$c_t(1 + \tau_c) + a_{t+1} = ws_t(1 - \tau_w) + a_t(1 + r(1 - \tau_r)) + \tau$$

where τ_c is cons. tax, τ_w labor income tax, τ_r capital income tax, and τ lump-sum transfer.

- Aggregate tax revenue is the sum of all taxes levied on the households.

$$T = \int \tau_w ws \lambda(a, s) + \int \tau_r ra \lambda(a, s) + \int \tau_c c \lambda(a, s) + \int \tau \lambda(a, s)$$

$$T = \tau_w wN + \tau_r rA + \tau_c C + \tau$$

- One tax instrument **must be chosen** so the government budget constraint is satisfied. All the others can be calibrated.

Extension: Fiscal Policy

- Must calibrate fiscal policy rules:
 - ▶ Fraction of gov. expenditure of GDP: $g_y \equiv G/Y$.
 - ▶ Public debt-to-GDP: $b_y \equiv B/Y$.
- What is the effect of higher public debt? Aiyagari and McGrattan (1998, JME) study what is the **optimal government debt level** (i.e., b_y).
 - ▶ Some debt may be good since it provides liquidity for the HH and raises r .
 - ▶ But distortionary taxation is bad and G crowds out investment.
 - ▶ They find that some debt is welfare improving, but the effects are small
- When considering life-cycle motives, Peterman and Sager (2022, AEJ: Macro) find that **public savings** is optimal.

Extension: Progressive Taxation

- A functional form that captures progressivity (See Benabou (2002), Heathcote et al. (2017)):

$$T(y) = y - \tau_1 y^{1-\tau_2} \quad \text{where } y \text{ is the individual labor income.}$$

- ▶ τ_2 gives the degree of progressivity, i.e. it measures the elasticity of post-tax to pre-tax income.
 - ▶ Given τ_2 , τ_1 shifts the tax function and determines the average level of taxation in the economy.
- Aggregate tax revenue is the sum (integral) of all individuals in the economy:

$$T = \int T(y_i) di$$

- ▶ Gov. budget can be balanced either by shifting the fraction of gov. expenditure, g_y , (as in Heathcote et al (2017)), or by adding an extra lump-sum transfer (as in Boar and Midrigan (2022)).

Extension: Progressive Taxation ($T(y) = y - \tau_1 y^{1-\tau_2}$)

- The tax is progressive if the ratio of marginal to average tax rates is larger than 1 for every level of income.
 - ▶ $\tau_2 = 1$: full redistribution $\Rightarrow T(y) = y - \tau_1$.
 - ▶ $0 < \tau_2 < 1$: progressivity $\Rightarrow T'(y) > \frac{T(y)}{y}$.
 - ▶ $\tau_2 = 0$: no redistribution $\Rightarrow T'(y) = \frac{T(y)}{y} = 1 - \tau_1$.
 - ▶ $\tau_2 < 0$: regressivity $\Rightarrow T'(y) < \frac{T(y)}{y}$.
- Break-even income: $y_{be} = \tau_1^{\frac{1}{\tau_2}}$.
 - ▶ If $y_i > y_{be}$, i is a taxpayer.
 - ▶ If $y_i < y_{be}$, i receives a transfer.

How to Evaluate Optimal Policy?

- Suppose we want to evaluate two tax levels (τ_0 or τ_1).
 - ▶ **Representative Agent:** Compare differences in utility of the RA.
 - ▶ **Heterogeneous Agent:** There is a distribution of welfare. Must specify a Social Welfare Function.
- The most common is Utilitarian. See Boar and Midrigan (2022) and Bénabou (2002) for a discussion.
- We have to compute the average lifetime utility weighted by the distribution for both policies:

$$W(\tau) = \int_{A \times S} V(a, s; \tau) d\lambda$$

where $V(a, s; \tau)$ expected lifetime utility for policy τ :

$$V(a, s; \tau) = \mathbb{E} \sum_{t=0}^{\infty} \beta^t \frac{(c_t)^{1-\gamma}}{1-\gamma} \quad \text{s.t.} \quad \text{Budget Constraint}$$

How to Evaluate Optimal Policy?

- Comparing different policies: we must take into account risk, endogenous distribution, curvature of utility, etc $\Rightarrow$ use **Consumption-equivalent variation (CEV)**.
- **CEV** $\Rightarrow$ $\% \Delta$ by which every HH consumption has to be changed in order to make it indifferent between the two policies: $W(\tau_0) = W(\tau_1, \Delta)$, where:

$$W(\tau_1, \Delta) = \int_{A \times S} \mathbb{E} \sum_{t=0}^{\infty} \beta^t \frac{(c_t^*(1 + \Delta))^{1-\gamma}}{1 - \gamma} d\lambda =$$

$$W(\tau_1, \Delta) = (1 + \Delta)^{1-\gamma} \int_{A \times S} \mathbb{E} \sum_{t=0}^{\infty} \beta^t \frac{(c_t^*)^{1-\gamma}}{1 - \gamma} d\lambda = (1 + \Delta)^{1-\gamma} W(\tau_1)$$

- If $\Delta > 0$, then avg. welfare is higher in policy τ_0 :

$$W(\tau_0) = W(\tau_1, \Delta) \quad \Leftrightarrow \quad \Delta = \left(\frac{W(\tau_0)}{W(\tau_1)} \right)^{1/(1-\gamma)} - 1$$